

F1R3SCORE · WORKING NOTE · SEPTEMBER 2026

# Scores as processes

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A graded process calculus for composition and algorithmic music:  
notes as communication, idioms as formulae, style and physicality  
as grading.

L. G. Meredith · F1R3FLY.io



# F1R3FLY.IO

## WHO THIS DECK IS FOR

# Two readers, one notation

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### FOR THE MUSICIAN

#### **A score that says who listens to whom**

The communication between players, the preferences of a style and what lies under the hands become written parts of the score. A performance is one path through it.

### FOR THE AI RESEARCHER

#### **A worked example of graded where-clauses**

Every modelling choice is a choice in the rho calculus: notes are communication events, style is a product of graded factors, idioms are spatial and behavioural formulae. The claims are theorems and measurements.

Slides 3–16 build the musical picture. Slides 17–21 read the same material as a template for applying graded rho to other domains.



# What a page of notation leaves to the player

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Conventional notation is a very good description of one performance.

Where the music is itself a structure of exchange, as in jazz or across many African musics, the grid records only the result.

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## **Which note a line would rather go to**

A style is a set of preferences over continuations, not a single written line.

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## **What a phrase is answering**

Call and response, trading fours and interlocking parts are structures of exchange.

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## **What a hand can reach**

Players follow their hands. Ease of playing shapes every idiom.



# Three things the score writes down explicitly

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1

## Who plays with whom

Players are parts of the process.  
Handing a line to another player, or  
alternating the notes of one line  
between two players, is a  
communication.

2

## How likely each continuation is

Every choice weighs its alternatives.  
Factors for style, register and  
instrument multiply; the normalised  
product is what plays.

3

## What the music must or should do

Formulae read the recent past (motifs)  
and look ahead in the company of the  
other players (idioms). Rules forbid;  
weights encourage.

A score written this way is a distribution over performances.



## FOR THE MUSICIAN

# A note is where a pitch meets a duration

### THE HAND · A CHANNEL

```
for (_ ← <@L, piano, C4>)  
{ ...next... }
```

Holds a pitch and waits.



**C4 · quarter**

one note, emitted



### THE MESSAGE · A CO-CHANNEL

```
<@L, piano, q>!(...)
```

Offers a duration.

They meet only when their locations agree: both are at @L. A pitch never meets a pitch.

Every meeting emits exactly one note. There are no silent steps anywhere in the calculus.

A rest is a pitch like any other. A rest of length zero is how the machinery of a score stays inaudible.



# Composition supplies the musical vocabulary

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## Harmony

### NO SHARED LOCATION

The parts are independent. Their events commute, and notes at equal musical times sound together.

## Melody

### A LOCATION, ONE WAY TO MEET

Interaction. Each note sets up the next through its continuation.

## Choice

### A LOCATION, SEVERAL WAYS TO MEET

Contention. Some candidates must lose, and this is where the weights live.

Melody and harmony both come from putting parts side by side, written  $P \mid Q$ . They differ only in whether the parts interact or stay independent.

## FOR THE MUSICIAN

# Twinkle, Twinkle, Little Star as a score

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```
score Twinkle
import std
pitches    { r, C3, F3, G3, C4, D4, E4, F4, G4, A4 }
durations { q = 1/4, h = 1/2 }
timbres    { piano = gm(0) on 1, bass = gm(32) on 2 }

play line(base "Lm", piano)
    [ C4 q, C4 q, G4 q, G4 q, A4 q, A4 q, G4 h,
      F4 q, F4 q, E4 q, E4 q, D4 q, D4 q, C4 h ]
| line(base "Lb", bass)
    [ C3 h, C3 h, F3 h, C3 h, F3 h, C3 h, G3 h, C3 h ]
```

**line** is a chain of written notes. Each note's continuation brings the next hand and message forward.

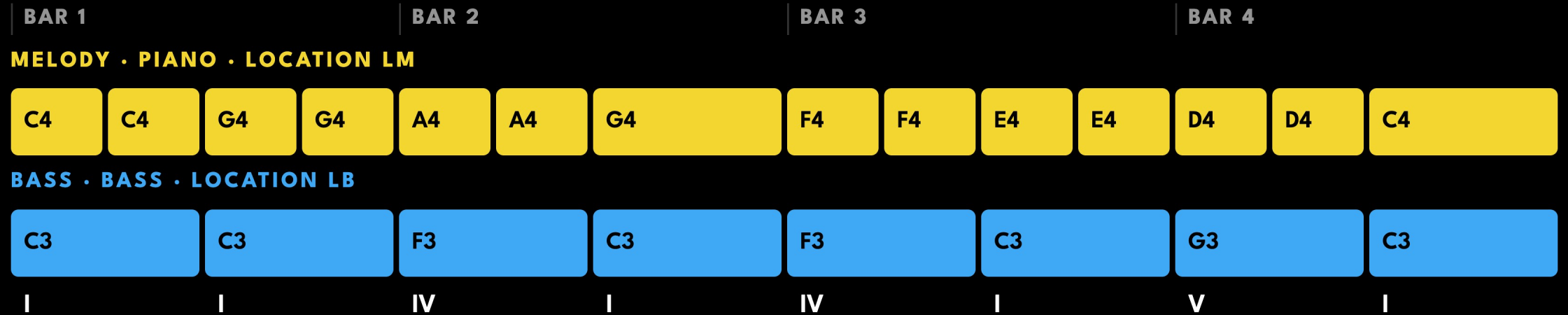
The two parts sit at different locations, Lm and Lb, in different timbres. Nothing in the text coordinates them.

It is a finite term: no recursion, no where-clauses, no grading. A fixed piece is the simplest case of everything that follows.



## FOR THE MUSICIAN

# Harmony without a conductor

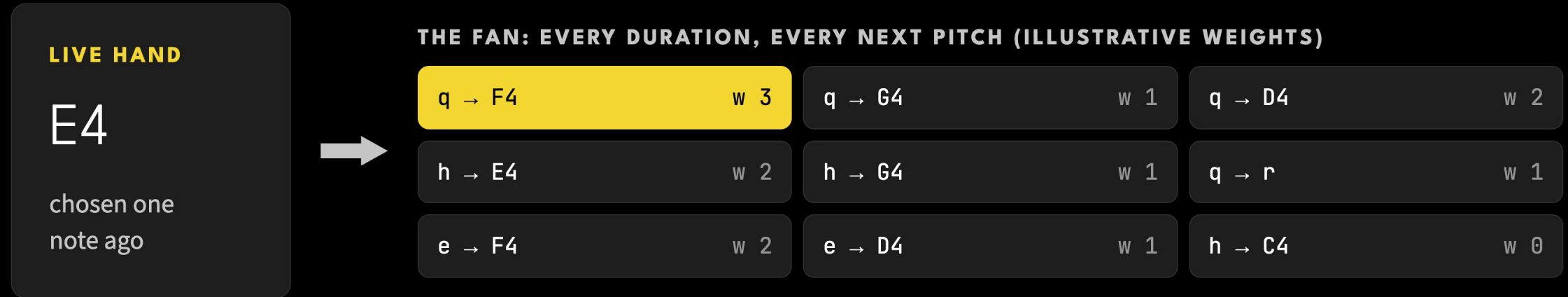


Each part keeps its own clock in its own durations, and the notes line up because the durations add up. Every onset-ordered run plays the same 22 timed notes and ends at four whole notes; the eight shared onsets are the chords of the phrase.



## FOR THE MUSICIAN

# A voice: a hand, a fan, and its own past



The hand already holds its pitch. The fitting chooses how long it lasts and which pitch comes next.

The chosen record becomes the next location, so a voice's location is its past, most recent note outermost.

Everything not chosen is left where no receipt will ever look. The unplayed is inert.

## FOR THE MUSICIAN

# Style and physicality are separate factors

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```
factor stepwise = table step { -2..2: 3, _: 3/10 }
factor country  = machine pitch { C4 → D4 @ 2,
                                   C4 → E4 @ 3, ... }
factor rhythm   = machine dur uniform

play Voice(piano, country * stepwise * rhythm,
           C4, q, "A")
  | Voice(bass, walking * rhythm,
         C3, h, "B")
```

### Product \*

All preferences at once. A continuation is likely only if every factor accepts it, and any factor can veto with 0.

### Sum +

A blend of styles. Scaling one summand sets the mix.

### Interval factors

Weights that depend only on the step make the line a sum of intervals, so combining them convolves interval laws.

Only ratios are heard: a composer writes weights on whatever scale is convenient.



## FOR THE MUSICIAN

# One style, two instruments

A fixed style over twenty-one pitches is multiplied first by a stepwise instrument (steps of at most two degrees weighted 3, all else 0.3), then by a leaping one (intervals of four or seven degrees weighted 3, all else 0.3).

| measured over 20,000 notes each         | stepwise instrument | leaping instrument |
|-----------------------------------------|---------------------|--------------------|
| share of steps of at most two degrees   | 0.582               | 0.135              |
| share of leaps of four or seven degrees | 0.051               | 0.378              |
| mean interval, in degrees               | 4.91                | 6.97               |

In both cases the played transitions match the normalised product, and no transition the product forbids ever occurs. Imagined physicalities, such as a hand spanning three octaves or strings tuned in thirds, are one-line changes to the score.

## FOR THE MUSICIAN

# Motifs and rules read the voice's past

### ENCOURAGE · A GRADED FACTOR

#### Complete the motif 0, 2, 4

```
[ last[0, 2] && carry(4) ] * 20 + 1
```

The head 0, 2 is completed by 4 at a rate that rises from 0.171 to 0.775 over 10,000 notes, against analytic values of 0.154 and 0.792.

### ENFORCE · A CRISP FACTOR

#### No pitch three times in a row

Unconstrained, a line of 10,000 notes contains 1,303 threefold repeated pitches. With a crisp factor forbidding a pitch equal to the last two, it contains none, and the voice never falls silent.

Transformations of a motif are formulae too: an answer transposed by a fixed interval, an inversion, or an echo of the rhythm with new pitches.



## FOR THE MUSICIAN

# Improvisation is a trace through the score

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### THE CRISP LAYER

What may be played: the tune, the changes, the rules of the idiom. A crisp factor can only shrink the set of admissible traces.

### THE GRADED LAYER

What the idiom would rather play. Changing positive weights changes how likely a trace is, never whether it is admissible.

### Algorithmic

samples a trace with the probabilities the grading gives.

### Human improviser

chooses freely among the candidates of positive weight.

### Mixed

an accompanist driven by the score follows a human soloist.

With a viability factor, a performer who stays within the score can never be left with nothing admissible to play.



## FOR THE MUSICIAN

# Exchange between players is written directly

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### Call and response

A caller hands the line to a responder. The handoff is itself a note, and it carries the caller's whole history, so the response can read the call: answer a fifth higher, or echo its rhythm.

### Hocket and interlocking

Two players share one line, each serving the other's next note. The line's factors see every note while each player adds their own preferences, as with players sharing an amadinda xylophone.

### Cross-rhythm

Three against two is two independent voices, one using a triplet value of  $\frac{1}{6}$  and one a value of  $\frac{1}{4}$ . Onsets coincide at every half note whatever pitches are chosen, and each part keeps its own pulse.



## FOR THE MUSICIAN

# Free harmony, coupled voicing

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### INDEPENDENT VOICES

#### **Product law, same performance under any schedule**

Five schedulers over 3,000 resolutions produced identical performances, with 216 simultaneous strikes. Nothing coordinates the voices; each keeps its own clock.

### CHORDS AS JOINS

#### **One receipt that weighs a whole voicing**

A chord receipt waits for several durations at once and sees every tone of the next chord together. That is where vertical physicality lives: closed stacked thirds under a pianist's hand, open fifths on strings tuned in fifths.

A hand is not a set of independent fingers. The calculus keeps harmony and voicing apart, and the composer chooses which relation holds between which parts.



## FOR THE MUSICIAN

# Where the idea came from: F1R3Skein

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F1R3Skein is a virtual instrument that fits a pitch stream and a duration stream into notes: first from the digits of transcendental constants, later from weighted state machines the player draws.

Two streams meeting at a note suggested channels and co-channels. The aims differ: the instrument is for a performer, and the calculus is a notation for a composer. The instrument is one score among many.

```
f1r3score play t4_embedding.score \  
  --notes 16 --gc freshness \  
  --stream A.pitch=spigot:pi/16@1 \  
  --stream A.dur=spigot:e/5
```

The pitches are the hexadecimal digits of  $\pi$  and the durations the base-5 digits of  $e$ . Over 400 notes the streams equal the digit streams, one digit per draw.



# Why this is worth an AI researcher's time

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A generative model of music in which every design choice is a choice in the rho calculus with graded where-clauses, and every claimed behaviour is a theorem or a measurement.

## Concrete

Alphabets are finite, weights are exact rationals, and a reference simulator and a Rust player implement the same semantics.

## Compositional

Style, physicality and idiom are separate graded factors with a proven law for their product.

## Constrained and probabilistic

Hard rules and soft preferences are different sorts of formula, and they provably do not interfere.



# Reading the calculus as a template

| in the calculus                                      | in the score                                 | the general pattern              |
|------------------------------------------------------|----------------------------------------------|----------------------------------|
| COMM between name-equivalent locations               | a note: pitch meets duration                 | each event a typed interaction   |
| weights normalised per contention set                | which continuation is played                 | weighted, normalised rule choice |
| $\otimes$ of graded factors                          | style $\times$ register $\times$ physicality | product of experts; 0 is a veto  |
| crisp $[\beta]$ beside graded $\psi$                 | rule beside tendency                         | support apart from likelihood    |
| spatial atoms over the quoted location               | motifs read from the past                    | memory as a structured name      |
| $\langle K \rangle$ modalities with a $vX$ viability | an idiom kept in company                     | shielded generation, no deadlock |
| reflection: @P and $*x$                              | unbounded voices, silently                   | recursion by reflection alone    |



# What is proved rather than hoped for

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## Law of a voice

Each fitting is drawn from the normalised product of the clause's factors, conditioned on the past.

## Factorised resolution

Factors over disjoint aspects are heard independently and can use separate chance sources.

## Support and grading

Positive weights never change which traces are admissible; crisp factors only shrink the support.

## Rules without deadlock

With a viability greatest fixed point, a voice never breaks the rule and never falls silent.

## Free harmony

Independent voices have product law and the same performance under every interleaving.

## Two-sided fitting

A single note cannot choose from both sides at once; one datum is fixed a note early.



# The ideal defines; the checkable decides

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The viability formula is a greatest fixed point in the ideal logic, quantifying over the contexts the rest of the score supplies.

When a rule reads a bounded window of the past and the alphabets are finite, the relevant state is the window, the held pitch and the recorded duration. That is a finite graph, on which the fixed point is computed by iteration.

This is the ideal-versus-checkable discipline of the rho-life notes, applied to a working generator.

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## OPEN THREADS

### Learning styles

Paper II's gradient in the fibre can fit a style factor to a corpus; the composer's formulae say which features the learner may use.

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### Complex grading

Over  $\mathbb{C}$ , two phrasings reaching one configuration would interfere; a voice already has what the quantum fragment needs.

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### Tempo

A second clause, valued in a second resolution algebra, that scales durations as continuations are stamped.



# Checked by simulation, played by a Rust player

## skeinsim.py · reference simulator

### LAW OF A VOICE

20,000 notes; largest deviation from the pitch matrix 0.031 and from the duration matrix 0.009.

### RECURSION BY REFLECTION

identical to the step semantics over 3,000 notes; exactly two null notes per note, all silent.

### THE UNPLAYED IS INERT

$79 = 5 \cdot 16 - 1$  inert messages left per note over 400 notes; none ever joins a candidate.

## F1R3Score · the player

### FORMATS

a table, JSON, a replayable JSONL trace, a MIDI file, or live MIDI

### CORE

exact rationals, hash-consed terms in congruence normal form, maximal matchings, factor routing

### TESTS

48, including the ported suite T0–T12 and differential checks against skeinsim.py, fitting by fitting



## GETTING STARTED

# Where the work lives

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### F1R3FLY-IO/PUBLICATIONS · F1R3SCORE

Scores as Processes (score-calculus), the player specification, and skeinsim.py

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### F1R3FLY-IO/F1R3SCORE

the Rust player: check, expand, play, replay

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### F1R3FLY-IO/PUBLICATIONS · FUZZYWARE

the graded where-clause papers: Paper I (quantum) and Paper II (over the reals)

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### F1R3FLY-IO/F1R3GAMES · F1R3SKEIN

the virtual instrument the idea grew out of

## TRY IT

```
f1r3score check examples/twinkle.score
f1r3score play  examples/twinkle.score \
    --midi twinkle.mid
f1r3score play  examples/hocket.score \
    --notes 16 --chance human
```

The last command lets you choose each fitting yourself, as the improviser of the score.



A score is a distribution over performances, and a performance is a trace through it.

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Scores as Processes · F1R3FLY.io working note, September 2026



**F1R3FLY.IO**